Which genes can a pathology foundation model see?

Cellularity confounding and transcriptome-wide predictability from H&E

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Abstract

Predicting spatial transcriptomic profiles directly from routine hematoxylin and eosin (H&E) histology images is widely pursued to augment morphological pathology with molecular measurements. However, many benchmarks evaluate models on un-normalised count targets ($\log(1 + \text{count})$) and restricted gene panels, which can conflate expression prediction with the number of cells in a spot. Here, we present a systematic predictability atlas across ∼5,000 genes measured by Xenium Prime 5K on 9 human tissue samples spanning six organ sites (lung, ovary, cervix, skin, prostate, lymph node) and 8 leave-one-group-out folds. On raw counts, state-of-the-art pathology foundation models (Virchow2, UNI2-h, CONCH v1.5) achieve gene-averaged Pearson correlations up to $r = 0.519$, compared with $r = 0.445$ for the number of cells in each spot (taken from the Xenium cell segmentation, not from the image) and $r = 0.490$ for marker-based cell-type composition. After standard library-size normalisation (lognorm), performance falls to $r = 0.193$ (Virchow2), $r = 0.148$ (UNI2-h) and $r = 0.135$ (CONCH v1.5), against $r = 0.214$ for the cell-count baseline. In paired comparisons with a bootstrap over held-out groups, Virchow2 is statistically indistinguishable from the cell count (difference -0.021, 95% CI [-0.093, 0.027]) and exceeds cell-type composition, while UNI2-h and CONCH v1.5 fall below the cell count. Gene-level features (expression level, sparsity, spatial autocorrelation, cell-type specificity, cross-encoder agreement) jointly explain $R^2 = 0.953$ of the variation in per-gene r on raw counts and $R^2 = 0.610$ on normalised counts. Our results suggest that much of the apparent predictability of raw counts from histology reflects how many cells a spot contains, and that baselines of this kind should be reported alongside image models.

1 Introduction

Spatial transcriptomics enables spatially resolved molecular characterisation of complex tissues, offering profound insights into tumor heterogeneity and microenvironmental architecture. However, experimental spatial profiling remains cost-prohibitive for high-throughput clinical application. In contrast, routine hematoxylin and eosin (H&E) staining is universal, rapid, and economical. This contrast has motivated a growing literature aiming to predict spatial gene expression directly from H&E morphology using deep learning [4, 10, 12, 5] and pathology foundation models [6, 1, 11].

Reported accuracies vary widely across studies and datasets. We focus on three aspects of evaluation that, to our knowledge, are rarely examined together:

1. **Target confounding:** Many evaluations operate on raw or un-normalised log-transformed counts ($\log(1 + \text{count})$). In multi-cellular spots or pseudo-spots, total read count is strongly governed by local cell density (cellularity), a well-known technical confound in spatial biology [3, 7].

2. **Restricted gene subsets:** Benchmarks often report performance on selected panels of 50 highly variable genes, which over-indexes on high-abundance lineage markers and obscures transcriptome-wide behavior.

3. **Cellularity baselines:** To our knowledge, few studies compare foundation models against trivial non-imaging baselines, such as scalar spot cell counts (n_cells) or broad cellular composition.

We construct a predictability atlas across 9 human Xenium Prime 5K samples spanning 6 organ systems, evaluating 4,624 shared genes across 8 leave-one-group-out (LOGO) test folds. We evaluate both un-normalised (log1p) and library-size normalised (lognorm) targets, bench-marking foundation models against non-imaging scalar cellularity and single-cell composition baselines. We find that a per-spot cell count accounts for much of the raw-count predictability, and that after normalisation the best foundation model is on par with that baseline while the others fall below it.

2 Related Work

Spatial Gene Expression Prediction from Histology. Early approaches to predicting spatial gene expression from H&E histology images, notably ST-Net [4], used convolutional architectures to regress spot-level log-transformed counts on 10x Visium breast cancer sections.

Subsequent methods introduced Transformer-based architectures to incorporate spatial context across neighbouring spots (HisToGene; [10]) and bi-modal contrastive representations aligning image patches with transcriptomic profiles (BLEEP; [12]). Recent translational studies such as DeepPT [5] have applied imputed transcriptomes from whole-slide images to predict downstream clinical therapy response. Many of these studies evaluate on log-transformed counts ($\log(1+\text{count})$); to our knowledge, explicit cell-count baselines are rarely reported.

Pathology Vision Foundation Models. Recent years have seen the development of large-scale self-supervised foundation models pretrained on hundreds of millions to billions of histopathology image patches, including UNI and UNI2-h [1, 9], Virchow [11] and Virchow2 [13], and multimodal vision-language models such as CONCH [8] and TITAN [2]. The HEST-1k benchmark [6] systematically evaluated these models using linear probes on frozen representations across diverse spatial transcriptomics datasets, establishing foundation model representations as standard feature extractors for morphology-to-expression mapping.

The Cellularity Confound and Spatial Normalisation. In single-cell and spatial transcriptomics, total spot count is well recognized as a major technical confound reflecting sequencing depth and the total number of cells captured per spot [3, 7]. Deconvolution methods explicitly separate cell count from cell-type abundance. In this work, we compare frozen pathology foundation models against a per-spot cell-count baseline on both unnormalised and library-size-normalised targets.

3 Data and Methods

Datasets and Pseudo-Spots. We analyze 9 human Xenium Prime 5K samples from HEST-1k v1.3.0 [6], spanning six organs: lung (LUAD and PDL TMAs), ovary (FF and FFPE), cervix, skin (melanoma), prostate (PRAD), and reactive lymph node. (One sample, TENX190, is the 541-gene Xenium v1 run of the TENX189 block and is excluded). Samples from the same TMA study or tissue block are assigned to the same held-out group, yielding 8 leave-one-group-out (LOGO) test folds. For each sample, single-cell transcripts (QV ≥ 20) are mapped to high-resolution H&E coordinates and aggregated into pseudo-spots aligned with the 112 μm footprint of HEST image patches (224 px at 0.5 μm/px). All 9 samples share 4,624 common genes.

Foundation Models and Probe Protocol. We extract frozen patch embeddings from state-of-the-art pathology foundation models: UNI2-h (SwiGLU ViT-Giant, 1536-dim), Virchow2 (ViT-Giant, class token + average patch token, 2560-dim), and CONCH v1.5 (ViT-B, 768-dim). Following the HEST protocol, features are standardized and reduced to 256 dimensions via exact incremental stream PCA. Per-gene ridge regression heads are fitted with penalty parameter α tuned via inner-group cross-validation strictly within training samples.

Non-Imaging Baselines. To decouple biological morphology from simple cellularity, we construct four baselines:

1. **Cell count (*n_cells*):** The number of cells within each spot's footprint, taken from the Xenium cell segmentation (molecular data, not the H&E image). It is an oracle-style baseline: it uses information an image model must infer.
2. **Coarse composition:** Cell-type proportions derived from standard lineage marker expression.
3. **Unsupervised single-cell clustering:** To prevent marker bias, single cells are clustered unsupervised (log-normalisation, PCA-50, kNN graph, Leiden community detection). Critically, clustering is fitted *strictly on training samples per fold*, and test cells are projected onto training centroids (0.0% unassigned cells; a unit test checks that test samples are not used to fit the clustering).
4. **Combined baseline:** Feature concatenation of cell count and composition.

Evaluation Metrics. Models are evaluated by per-gene Pearson correlation r on held-out test groups, averaged over genes and folds. We evaluate both raw $\log(1+\text{count})$ and library-size normalised (lognorm) targets. A permutation control shuffles spot expression relative to image embeddings and gives $r = -0.000$, consistent with no leakage. Confidence intervals and paired comparisons use a bootstrap over held-out groups; because genes are not independent, we do not report per-gene significance tests.

4 Results

4.1 Reproducing HEST-Benchmark

Table 1 validates our pipeline against published HEST-Benchmark numbers, reproducing performance across IDC, LUNG, and COAD with $|\Delta| \leq 0.001$.

4.2 A Transcriptome-Wide Predictability Atlas

Table 2 reports predictability across all 4624 genes. On raw counts, Virchow2 reaches $r = 0.519$, followed by UNI2-h ($r = 0.479$). A per-spot cell count alone achieves

Task	Encoder	Ours	Published	Δ
IDC	ResNet50	0.474	0.474	-0.000
IDC	UNI2-h	0.590	0.590	0.000
LUNG	ResNet50	0.495	0.495	0.000
LUNG	UNI2-h	0.559	0.559	0.001
COAD	ResNet50	0.250	0.250	0.000
COAD	UNI2-h	0.303	0.301	0.001

Table 1: HEST-Benchmark reproduction (ridge + PCA-256, published protocol).

Model	mean r	95% CI	median r	% r>0.3
UNI2-h	0.479	[0.397, 0.548]	0.519	73.9
Virchow2	0.519	[0.444, 0.574]	0.566	75.7
CONCH v1.5	0.474	[0.393, 0.536]	0.510	71.1
Composition (oracle)	0.490	[0.424, 0.548]	0.529	73.5
Cell count	0.445	[0.354, 0.528]	0.480	72.3
Composition + cell count	0.503	[0.432, 0.566]	0.546	74.7

Table 2: Xenium 5k atlas, leave-one-group-out, 4624 genes.

$r = 0.445$ without any image features; Virchow2 exceeds it by 0.075 (95% CI [0.027, 0.121]).

Under library-size normalisation (lognorm), gene-averaged r falls by more than 60% for all models: Virchow2 to $r = 0.193$, UNI2-h to $r = 0.148$ and CONCH v1.5 to $r = 0.135$, against $r = 0.214$ for the cell count and $r = 0.153$ for composition. Paired over held-out groups, Virchow2 is on par with the cell count (difference -0.021, 95% CI [-0.093, 0.027]) and above composition (0.040, [0.018, 0.064]), whereas UNI2-h (-0.066, [-0.148, -0.006]) and CONCH v1.5 (-0.079, [-0.156, -0.025]) fall below the cell count.

4.3 Unsupervised Single-Cell Clustering Baseline

To determine whether coarse marker cell typing artificially deflated the composition baseline (due to 72.3% unassigned cells), we evaluated the unsupervised Leiden clustering baseline. Despite categorizing 100% of cells into data-driven clusters (0.000 unassigned cells), cluster composition reached $r = 0.357$ on raw counts and $r = 0.207$ on normalised counts. On normalised counts this does not exceed the cell count ($r = 0.223$), suggesting that the weak composition baseline was not caused by unassigned cells alone.

4.4 What Explains Residual Predictability?

On raw counts, spatial autocorrelation (Moran's I), sparsity, and expression level together with the other gene-level features jointly explain $R^2 = 0.953$ of cross-gene variance in r (Spearman ρ with Moran's I is 0.946). On normalised targets (lognorm), cross-gene variance is explained with $R^2 = 0.610$, dominated by cell-type specificity (τ) and cross-encoder agreement rather than total expression. Some genes are predicted well above both baselines; for example, the fibroblast-associated genes CTHRC1, THY1 and ADAM12 (Virchow2, normalised targets), which may reflect morphologically visible stromal states.

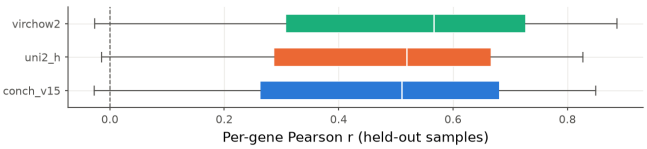


Figure 1: Per-gene Pearson r on held-out groups.

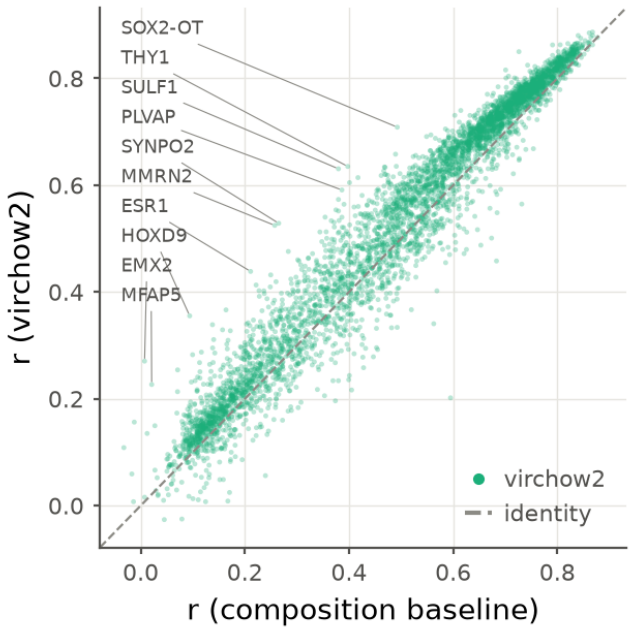


Figure 2: Image model vs. composition baseline per gene.

4.5 Spatial Context

Adding the mean representation of the first ring of neighbouring patches changes gene-averaged r by 0.004 on raw counts and 0.010 under library-size normalisation (160 genes gain > 0.05). Attention pooling over spatial neighbours provides no benefit over uniform ring averaging (-0.027).

4.6 Cross-Platform Generalisation

The analyses in this subsection were run on the original six Xenium samples (5,001 shared genes, five held-out groups), before the three additional samples were added; they were not repeated on the full set. Table 3 evaluates cross-platform transfer from 10x Visium to Xenium pseudo-spots. Training on lung Visium samples yields $r = 0.132$ for Virchow2, a modest drop of -0.021 rela-

Training set	Encoder	mean r	Δ vs LOGO	n
visium same organ	UNI2-h	0.117	-0.026	3
visium same organ	Virchow2	0.132	-0.021	3
xenium logo	UNI2-h	0.148	0.000	6
xenium logo	Virchow2	0.164	0.000	6
xenium same organ	UNI2-h	0.173	0.019	5
xenium same organ	Virchow2	0.205	0.034	5

Table 3: Per-test-sample r by training set (library-size-normalised targets).

tive to within-platform LOGO ($r = 0.164$). Restricting training to same-organ Xenium samples yields $r = 0.205$, reflecting clear organ-specific signatures.

4.7 Robustness and Determinism

The analyses in this subsection were run on the original six Xenium samples (5,001 shared genes, five held-out groups), before the three additional samples were added; they were not repeated on the full set. Varying random seeds produces identical results across seeds 0, 1, and 2 (s.d. $< 10^{-12}$ for Virchow2, UNI2-h, and CONCH v1.5); because exact PCA ridge regression is strictly deterministic, seed variation does not serve as meaningful evidence of algorithmic robustness.

For pseudo-spot aggregation windows under normalisation (lognorm, Virchow2), 56/112/224 μm windows achieve $r = 0.143$ / 0.167 / 0.151 (peaking at the nominal 112 μm patch footprint). Restricting image field of view from 112 to 56 μm drops r from 0.148 to 0.117 under lognorm. Linear ridge matches a 1-hidden-layer MLP ($r = 0.477$ vs 0.484).

5 Discussion and Limitations

Our results suggest caution when interpreting correlations from histology-to-expression models. On unnormalised counts, a large part of the signal can be obtained from the number of cells per spot, so high correlation alone does not show that a model has learned gene-specific programs. After library-size normalisation, foundation models give modest correlations; the best model is on par with a cell-count baseline that uses molecular segmentation, and gene-level exceptions exist. We recommend reporting normalised targets and cell-count baselines alongside image models.

Limitations. Our study is constrained by the small number of publicly available Xenium 5K datasets with registered H&E (9 samples across 6 tissue types). Lack of explicit patient identifiers necessitates conservative group-level splits. The cross-platform and robustness analyses (Sections on generalisation and robustness) use the original six-sample set. Cross-platform transfer was evaluated on lung tissue only, and pseudo-spot aggregation footprints overlap due to grid sampling. The cell-count baseline uses Xenium segmentation, so it is an upper reference rather than an image-only competitor; an H&E nuclei-count baseline was not evaluated. We used linear probes on frozen embeddings following the HEST protocol; fine-tuned models may behave differently.

Data Availability

The spatial transcriptomics datasets analyzed in this study derive from HEST-1k [6] and are available under the CC BY-NC-SA 4.0 license. The precomputed per-gene predictability atlases and summary statistics generated in this work are publicly shared on Hugging Face at `https://huggingface.co/datasets/raju-ai/hest-xenium5k-predictability-atlas`.

Code Availability

The full evaluation pipeline and analysis scripts are available on GitHub at `https://github.com/raju-sah/histology-gene-expression-prediction`.

AI-Use Statement

The analysis code, experiments and manuscript draft were produced with AI coding agents (Anthropic Claude and Google Gemini) under the author's direction. The author reviewed the results, reported numbers and citations and takes full responsibility for the content.

Author Contributions

R.S. conceived the study, directed the analysis, reviewed the results and wrote the paper.

Funding

This work received no specific funding.

Competing Interests

The author declares no competing interests.

Acknowledgements and Licences

We thank the authors of HEST-1k [6] and the developers of UNI, UNI2-h, Virchow, Virchow2, and CONCH. Model weights were accessed under their respective non-commercial research licenses (CC BY-NC-ND 4.0 and CC BY-NC-SA 4.0) solely for frozen feature extraction. No

proprietary model weights, whole-slide image tiles, or patient identifiers are redistributed.

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